

TABLEAU – JOINS AND FUNCTIONS

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What are Joins?

- In Tableau, joins are used to merge data from different tables or datasets based on common fields or keys.
- These joins are fundamental when working with relational databases or multiple data sources to create a comprehensive dataset for analysis and visualization



Inner Join

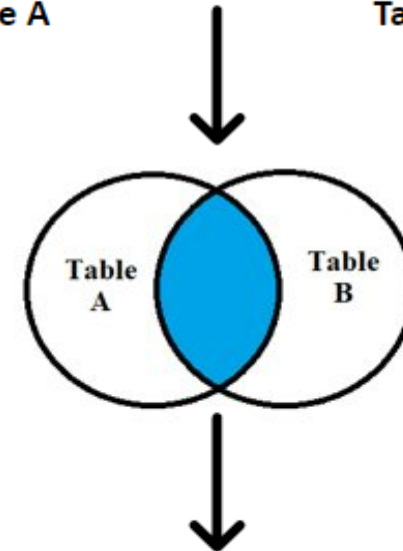
- This type of join returns only the rows where there is a match in both tables being joined.
- It includes records that have matching values in both tables based on the specified join condition.

Student ID	Name
1001	A
1002	B
1003	C
1004	D

Student ID	Department
1004	Mathematics
1005	Mathematics
1006	History
1007	Physics
1008	Computer Science

Table A

Table B



Student ID	Name	Department
1004	D	Mathematics

Left Join

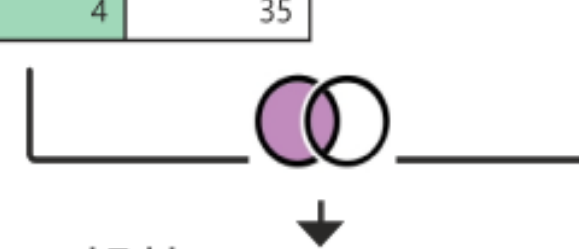
- A left join returns all records from the left table (the primary table) and the matched records from the right table.
- If there is no match in the right table, it returns NULL values for the right table fields.

Left Table

Date	CountryID	Units
1/1/2020	1	40
1/2/2020	1	25
1/3/2020	3	30
1/4/2020	4	35

Right Table

ID	Country
1	USA
2	Canada
3	Panama



Merged Table

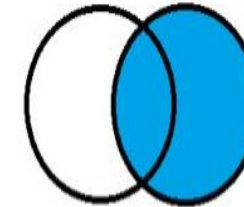
Date	CountryID	Units	Country
1/1/2020	1	40	USA
1/2/2020	1	25	USA
1/3/2020	3	30	Panama
1/4/2020	4	35	null

Right Join

- Similar to the left join, but it returns all records from the right table and the matched records from the left table.
- If there is no match in the left table, it returns NULL values for the left table fields.

Student ID	Name
1001	A
1002	B
1003	C
1004	D

Student ID	Department
1004	Mathematics
1005	Mathematics
1006	History
1007	Physics
1008	Computer Science



Student ID	Name	Department
1004	D	Mathematics
1005	NULL	Mathematics
1006	NULL	History
1007	NULL	Physics
1008	NULL	Computer Science

Full Outer Join

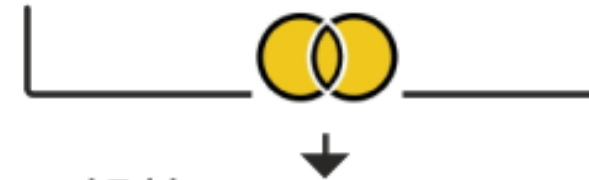
- This join type returns all records when there is a match in either the left or right table.
- It includes all rows from both tables and fills in NULL values where there is no match.

Left Table

Date	CountryID	Units
1/1/2020	1	40
1/2/2020	1	25
1/3/2020	3	30
1/4/2020	2	35

Right Table

ID	Country
1	USA
2	Canada
3	Panama
4	Spain



Merged Table

Date	CountryID	Units	Country
1/1/2020	1	40	USA
1/2/2020	1	25	USA
1/4/2020	2	35	Canada
1/3/2020	3	30	Panama
null	null	null	Spain

Union

- Union is used when you have multiple tables or files with the same columns or fields, but different rows of data.
- Union is often used during the data preparation phase in Tableau Desktop, allowing users to prepare and clean up data before creating visualizations or building dashboards.

trade

	dt	sym	price	size
0	2015-01-01	`C	10	10
1	2015-01-02	`C	10.5	100
2	2015-01-03	`MS	260	15
3	2015-01-04	`C	11	200
4	2015-01-04	`DBK	35.6	55
5	2015-01-05	`AAPL	1,010	20
6	2015-01-06	`AAPL	1,020	300
7	2015-01-07	`MS	255	200
8	2015-01-07	`MS	254	400

fbTrades

	dt	sym	size	book
0	2015-01-02	`FB	1000	`A
1	2015-01-03	`FB	1000	`B
2	2015-01-05	`FB	1000	`A

trade uj fbTrades

	dt	sym	price	size	book
0	2015-01-01	`C	10	10	`
1	2015-01-02	`C	10.5	100	`
2	2015-01-03	`MS	260	15	`
3	2015-01-04	`C	11	200	`
4	2015-01-04	`DBK	35.6	55	`
5	2015-01-05	`AAPL	1,010	20	`
6	2015-01-06	`AAPL	1,020	300	`
7	2015-01-07	`MS	255	200	`
8	2015-01-07	`MS	254	400	`
9	2015-01-02	`FB		1000	`A
10	2015-01-03	`FB		1000	`B
11	2015-01-05	`FB		1000	`A

What are Generated Fields?

- Generated fields are dynamic calculations created within Tableau
- They are based on pre-defined formulas or conditions



Benefits of Using Generated Fields

- Provides flexibility to perform complex calculations on the fly
- Allows for immediate exploration and experimentation with data





Tableau Functions

- Tableau functions are built-in formulas and calculations that allow users to manipulate and analyze data in Tableau.
- They can be used to perform various tasks such as aggregating, transforming, and manipulating data to create meaningful visualizations.

Number Functions

Function	Description	Example	Syntax
ABS()	Returns the absolute value of a number	ABS(-10) returns 10	ABS([Number])
ROUND()	Rounds a number to a specified number of decimal places	ROUND(3.14159, 2) returns 3.14	ROUND([Number], num_decimals)
SUM()	Calculates the sum of values in a field	SUM([Sales])	SUM([Field])
MAX()	Returns the maximum value in a field	MAX([Profit])	MAX([Field])
MIN()	Returns the minimum value in a field	MIN([Quantity])	MIN([Field])

String Functions

Function	Description	Example	Syntax
LEFT()	Returns the specified number of characters from the start of a string	LEFT('Tableau', 4) returns 'Tabl'	LEFT([String], num_characters)
RIGHT()	Returns the specified number of characters from the end of a string	RIGHT('Analytics', 4) returns 'tics'	RIGHT([String], num_characters)
LEN()	Returns the length of a string	LEN('Data') returns 4	LEN([String])
CONCAT()	Concatenates strings together	CONCAT('Hello ', 'World') returns 'Hello World'	CONCAT([String1], [String2])

Date Functions

Function	Description	Example	Syntax
YEAR()	Extracts the year from a date	YEAR(#2023-11-29#) returns 2023	YEAR([Date])
MONTH()	Extracts the month from a date	MONTH(#2023-11-29#) returns 11	MONTH([Date])
DAY()	Extracts the day from a date	DAY(#2023-11-29#) returns 29	DAY([Date])
DATEADD()	Adds or subtracts time from a date	DATEADD('month', 3, #2023-11-29#) returns 2024-02-29	DATEADD('interval', value, [Date])

Type Conversion Functions

Function	Description	Example	Syntax
INT()	Converts a value to an integer	INT(5.7) returns 5	INT([Value])
FLOAT()	Converts a value to a floating-point number	FLOAT('3.14') returns 3.14	FLOAT([Value])
STR()	Converts a value to a string	STR(123) returns '123'	STR([Value])
DATETIME()	Converts a string to a date	DATETIME('2023-11-29') returns #2023-11-29#	DATETIME([String])

avg count
sum std dev
max

Aggregate Functions

- Aggregate functions perform calculations on a set of values and return a single value
- Example: SUM, AVG, MIN, MAX, COUNT, COUNTD, MEDIAN, STDEV, VAR

Logical Function

Function	Syntax	Example
IF-THEN-ELSE	IF <condition> THEN <value_if_true> ELSE <value_if_false> END	IF [Sales] > 1000 THEN 'High Sales' ELSE 'Low Sales' END
AND	AND(<condition1>, <condition2>, ...)	AND([Quantity] > 10, [Profit] > 0)
OR	OR(<condition1>, <condition2>, ...)	OR([Category] = 'Furniture', [Category] = 'Office Supplies')
NOT	NOT <condition>	NOT [Region] = 'East'

Thank You



Thank You



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Web Resources

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